Research Proposal for Exchange Programme at IIT Delhi

Submitted by:
Ayaan Salim Shaikh
Enrollment No:2021BCHE003

DEPARTMENT OF CHEMICAL ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY SRINAGAR
SRINAGAR, JAMMU & KASHMIR (INDIA) - 190006

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Research Proposal

A standardized approach for permeance assessment in Direct Contact Membrane Distillation(DCMD).

1.Introduction:

Membrane distillation (MD) is a rising method that combines thermal and membrane processes to separate substances based on the difference in vapor pressure between two streams.In this technique,a hydrophobic porous membrane physically divides the streams, allowing vapor to transfer from a hot feed to a cold permeate side[1].

Despite its potential, large-scale adoption of MD remains limited worldwide. The key lies in fabricating MD membranes that exhibit **high hydrophobicity, permeability, resistance to fouling, mechanical strength, low thermal conductivity, and a high liquid entry pressure (LEP).**

Approximately **97%** of the world's water reserves are found in oceans, with only about **3%** available as freshwater. Most of this freshwater exists in the form of glaciers and **ice caps (2.06%)**, with a smaller portion as **groundwater(0.90%)** and **surface or other freshwater resources (0.03%)**[2,3].

The most common MD setup is **Direct Contact Membrane Distillation (DCMD)**, known for its simplicity in design and operation[4].The efficiency of MD largely depends on the properties of the membranes utilized.Therefore,there's a growing focus on developing membranes tailored specifically for MD applications.Techniques such as electrospinning of polymeric membranes[5],deposition of photothermal materials on membranes[6],and hydrophobization of inorganic membranes are being explored for this purpose.

From the process standpoint, transport limitations, which govern the extent of temperature polarization (TP) in MD, can lead to erroneous estimates of membrane performance. This is because the driving force is defined by the difference of vapor pressures estimated at the membrane surface temperature, not the bulk.Fortunato et al.[8] showed that the resulting **flux increased** from **23 to 42 kg m^{-2} h^{-1}** when the **feed flow rate increased** from **15 to 45 L/h**, respectively. Moreover, because the driving force is the difference in vapor pressure across the membrane,which involves exponential functions of temperature, it is used to normalize the flux,instead of the temperature difference.These shortcomings drive the need for a standardized methodology to compare and report membrane performance,especially when novel materials and membranes are developed. Specifically, the flux must not be used as an indicator of the membrane's productivity. Instead, the permeance, which is the flux normalized by the driving force (vapor pressure difference), should be used.

In light of these complexities, it is clear that a simple and practical experimental method for permeance measurements of MD membranes is required.

1.1 Current advantages and applications:

MD offers several advantages, including its ability to use low-grade energy, operate under low pressure, and achieve high solute rejection rates[9][10].Due to these benefits,**MD** finds applications in various fields such as **desalination**[11]**,industrial wastewater treatment**[12], and **food processing**[13].

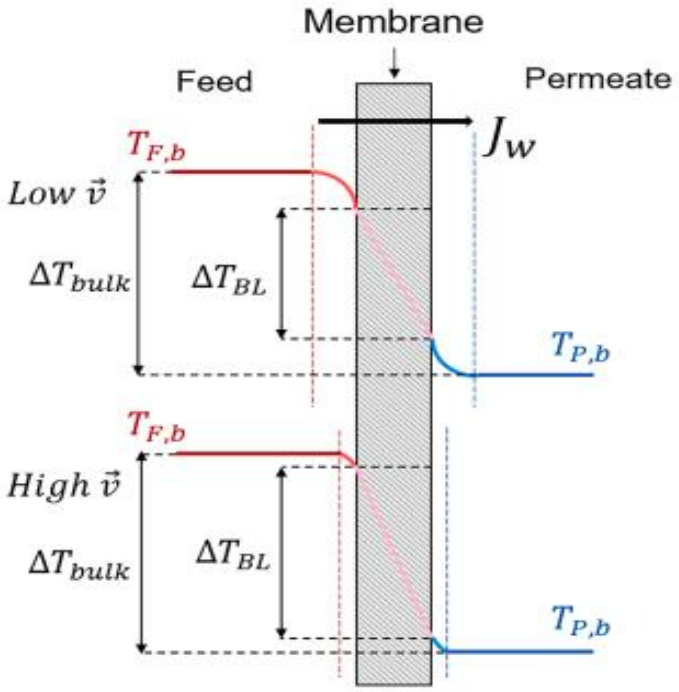

Fig. 1.1.1 The effect hydrodynamic conditions on temperature polarization and the driving force of a DCMD process at low cross-flow velocity (top) and high crossflow velocity (bottom)[14].

1.2 Current Research Problems:

1.2.1 Fouling and wetting in membranes:

In studies examining Membrane Distillation (MD) for solutions containing salts, the formation of scale on the membrane surface has been observed. Gryta [15] investigated MD performance in treating spent solutions from heparin production and reported a rapid decline in flux due to fouling and scaling. The presence of salt deposits on the distillate side confirms the occurrence of wetting.

1.2.2 Appropriate selection of Module length:

Longer modules yield **lower flux values**.Hence,the module length should be selected carefully such that it is long enough to yield stable measurements, but short enough to not influence the driving force.

2.Motivation:

2.1 Prospects and Challenges:

Membrane distillation (MD) holds promise as a separation technology across various industries, from desalination to pharmaceuticals and food processing. Despite its potential, MD's readiness for widespread adoption in desalination and water treatment markets remains constrained. One significant hurdle lies in the limited advancement and imprecise evaluation of MD-specific membranes**.** The complicated interaction between process parameters and membrane properties makes accurately gauging membrane performance challenging.

To address this, there's a pressing need for a standardized approach to assess and compare membrane performance,irrespective of process variables.

Based on the current methodology,work has been done on utilizing **a scaled-down Direct Contact Membrane Distillation(DCMD)**setup,

This scaled-down DCMD setup aims to **simplify membrane evaluation** and support current efforts in MD membrane advancement and process scaling.

2.2 Problem that I look forward to solve/considerations for future work:

Furthermore, while the current methodology can simplify and improve efforts in DCMD membrane development, there is a scope of further research to address key questions:

i)Determining an optimal membrane length for measuring permeance is crucial:

Since the driving force varies along the membrane length, influencing flux calculations, a standardized cell design with a shorter length becomes necessary for accurate membrane comparison.

ii)The potential issue of pore wetting at the maximum flow rate needs attention:

There is a probability that other membranes, whether commercial or novel, might encounter wetting before reaching the critical velocity to minimize the temperature polarization effect.

2.3 How can IIT Delhi help me?

Collaborating with faculty members and research staff at IIT Delhi with expertise in MD technology will provide valuable guidance and insights throughout the research process.

2.3.1 Access to Advanced Facilities:

IIT Delhi boast state-of-the-art facilities for membrane fabrication, characterization, and testing. Access to these facilities will enable the development and evaluation of novel MD membranes and experimental setups required for performance assessment.

2.4 Probable Labs:

At IIT Delhi:

1.**Membrane Science and Engineering Laboratory**: This laboratory could focus on the development, characterization, and testing of novel membrane materials and membrane distillation systems.

2**. Environmental Engineering Laboratory**: This laboratory might investigate the application of membrane distillation for water and wastewater treatment, including desalination and pollutant removal.

2.5 How these available instruments at IIT Delhi help me to uncover my research questions?

At IIT Delhi:

At IIT Delhi , the available instruments and facilities can greatly contribute to uncovering research questions on membrane distillation in several ways:

1.Membrane Fabrication Facilities: Facilities like spin coaters, electrospinning setups, and casting machines at IIT Delhi enable researchers to develop custom membrane materials for membrane distillation.

2.Surface Characterization Techniques: Advanced techniques like SEM, AFM, and contact angle goniometry at IIT Delhi analyze membrane surface properties,whose understanding is crucial for identifying fouling mechanisms and enhancing long-term stability in membrane distillation.

3.Research methodology:

Based on the current study methodology,I re-evaluated the following areas which could have a potential scope of further research:

a)To reduce the pore wetting in membranes.

b)Anti-fouling membranes.

c)To increase the lifespan/longetivity of membranes.

d)To increase the efficiency of membrane distillation(MD) process.

3.1 Probable research plan:

I try to intend to solve the various problems faced in the current study methodology by the following methods:

a)Pore wetting in membranes: To work on a methodology which could be further refined in future works to address potential pore wetting, which was not encountered in this work.

b)Membrane Pre-treatment:Pre-treating feed solutions before membrane filtration can remove or reduce foulants and prevent fouling and wetting. Techniques such as coagulation, flocculation, and microfiltration can be employed as pre-treatment steps.

c)Additives and Coatings: Incorporating additives or coatings into membranes can impart anti-fouling and anti-wetting properties.Examples include incorporating nanoparticles with anti-fouling properties or coating membranes with hydrophilic polymers or zeolite layers to repel foulants.

4.Conclusions:

Membrane distillation (MD) is a promising, energy-efficient separation process, utilizing low-grade waste or alternative energy sources like solar and geothermal energy. It operates at low temperatures (below 70°C) and can achieve high recovery factors and concentrations compared to conventional methods like reverse osmosis (RO). MD finds applications in various industries, including water treatment.

The current study shows that estimating based on flux depends on the process and doesn't accurately represent the membrane's performance.

Current study methodology also simplifies the evaluation of MD membranes in a way that's not tied to any specific process. It can help with developing MD membranes and scaling up processes.

5.References:

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